# Risk factors for multiple myeloma: a hospital-based case-control study in Northwest China.

BACKGROUND: The distinctive racial/ethnic and geographic distribution of multiple myeloma (MM) suggests that both family history and environmental factors may contribute to its development. METHODS: A hospital-based case-control study consisting of 220 confirmed MM cases and 220 individually matched patient controls, by sex, age and hospital was carried out at 5 major hospitals in Northwest China. A questionnaire was used to obtain information on demographics, family history, and the frequency of food items consumed. RESULTS: Based on multivariate analysis, a significant association between the risk of MM and family history of cancers in first degree relatives was observed (OR=4.03, 95% CI: 2.50-6.52). Fried food, cured/smoked food, black tea, and fish were not significantly associated with the risk of MM. Intake of shallot and garlic (OR=0.60, 95% CI: 0.43-0.85), soy food (OR=0.52, 95% CI: 0.36-0.75) and green tea (OR=0.38, 95% CI: 0.27-0.53) was significantly associated with a reduced risk of MM. In contrast, intake of brined vegetables and pickle was significantly associated with an increased risk (OR=2.03, 95% CI: 1.41-2.93). A more than multiplicative interaction on the decreased risk of MM was found between shallot/garlic and soy food. CONCLUSION: Our study in Northwest China found an increased risk of MM with a family history of cancer, a diet characterized by low consumption of garlic, green tea and soy foods, and high consumption of pickled vegetables. The effect of green tea in reducing the risk of MM is an interesting new finding which should be further confirmed. 